ToolBook 8 Architecture

Runtime flow, modules, and object model

High-Level Architecture

ToolBook is an interpreted, object-oriented, event-driven system built around a book metaphor. A .tbk file contains a PE stub, ZZZTBKBOOKLOC magic, and a JBO body of page records. TB80RTM.EXE is the engine: book loader, page table, message dispatch, and window classes (TBKMain, TBKClient, TBKChildWindow, SchedulerWndTB80). Supporting DLLs provide objects, painting, OpenScript, bitmaps, and type tables.

Key Modules (16-bit NE unless noted)

Module	Role
TB80RTM.EXE	Main engine, book open, windowing, message loop
TB80BAS.DLL	Objects, properties, painters, scale twips, type dispatch
TB80CMP.DLL	OpenScript bytecode, keyword pool, expand/emit
TB80BMP.DLL	Bitmaps, DIB, palette, graphic FX, LZ77
TB80UTL.DLL	Object type table (Page, Polygon, Field, …), helpers
TB80W16.DLL	Thunk layer, graphic IDs, script editor commands
nida81.tbk + .sbk	Shared shell, system books (base, recs, test, trnr, tb80r)

Book Open Path (partial)

tbload finds ZZZTBKBOOKLOC → JBO signature checked by virtual method → page structures allocated → CDBSENDCREATE with bounds → FUN_12c8_0850 creates object and writes bounds to object+0xF. The record-to-bounds extractor itself has not yet been located in the decompiles.

Painting Path

Live objects carry bounds at +0xF. Polygon painter (TB80BAS seg 0x82) reads count at +0x30 and point array at +0x32, scales twips via ScaleTwipsToScreen ((stored*96+720)/1440), then calls GDI!POLYGON. Arc painter (seg 0x7B) uses four twip pairs; rectangle painter reads bounds and calls GDI!RECTANGLE. Many shapes are computed from object bounds rather than stored point lists.

OpenScript

Handlers live in page records. Keyword tables are NE resources 0x77/0x78 in TB80CMP. Decoder recovers readable source (“go to page "P400"”, “send next”) including 0xC0–0xFF back-references from the compressed input stream.